



A randomized controlled trial to evaluate propylene glycol alone or in combination with dextrose as a treatment for hyperketonemia in dairy cows

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ABSTRACT

Our objective was to investigate the effect of i.v. dextrose as an adjunct therapy to oral propylene glycol on the resolution of hyperketonemia (HYK; blood β -hydroxybutyrate ≥ 1.2 mmol/L), disease incidence, and early lactation milk yield. Cows ($n = 1,249$) between 3 and 16 d in milk (DIM) from 4 New York dairy farms were screened once weekly for HYK for 2 wk. Those with HYK and no previous history of retained placenta, metritis, or HYK were randomly assigned to 1 of 3 treatment groups: 300 mL of oral 100% propylene glycol for 3 d (PG3); 300 mL of oral 100% propylene glycol for 3 d plus 500 mL i.v. 50% dextrose on d 1 (PG3D1); or 300 mL of oral 100% propylene glycol for 3 d plus 500 mL i.v. 50% dextrose on all 3 d (PG3D3). Cows with a blood β -hydroxybutyrate < 1.2 mmol/L at initial screening were re-screened the following week and randomly assigned to the above treatment groups if blood β -hydroxybutyrate was ≥ 1.2 mmol/L. Cows were assessed for post-treatment HYK resolution 1 and 2 wk after initial HYK diagnosis. We collected farm-diagnosed occurrence of adverse events (sold, died, metritis, displaced abomasum, or ketosis) during the first 60 DIM and milk yield data from the first 10 wk of lactation from herd management software. We used mixed-effects multivariable Poisson regression models to assess the risk of post-treatment HYK resolution at 1 and 2 wk following initial HYK diagnosis and adverse event occurrence among treated cows. We used repeated-measures ANOVA to assess differences in average daily milk yield between treatments. The overall HYK incidence was 30.1% ($n = 373$). Sixty-four percent of cows ($n = 237$) were assigned to a treatment group in the first week (3 to 9 DIM), and 36% ($n = 136$) assigned the second week (10 to 16 DIM). The incidence of 1 or more adverse events during the first

60 DIM was 9.4% ($n = 35$). We found no effect of treatment on risk of post-treatment HYK resolution at wk 1 (PG3 56.9%, PG3D1 45.0%, PG3D3 50.0%) or wk 2 (PG3 60.0%, PG3D1 52.1%, PG3D3 59.5%) following initial diagnosis, or for risk of adverse event occurrence (PG3 7.4%, PG3D1 8.0%, PG3D3 12.6%). Average daily milk yield (mean $\pm$ SE) was similar between treatment groups (PG3: 42.7 ± 0.6 kg/d, PG3D1: 42.4 ± 0.6 kg/d, PG3D3: 42.6 ± 0.6 kg/d). The addition of dextrose for 1 or 3 d provided no improvement in resolution of ketosis assessed once weekly, reduction in adverse events during the first 60 d of lactation, or a difference in average daily milk yield during the first 10 wk of lactation.

Key words: dairy cow, ketosis, treatment, dextrose, propylene glycol

INTRODUCTION

Dairy cows experience a period of nutrient deficit during early lactation as their rapidly increasing milk production outpaces their ability to increase DMI (Bauman and Currie, 1980; Baird, 1982; Herdt, 2000). As part of the homeorhetic adaptation to lactation, cows respond to this deficit by mobilizing adipose tissue through lipolysis. The associated increase in circulating fatty acid concentrations can overload the oxidative capacity of the liver, resulting in the formation of ketone bodies (e.g., BHB). Although ketones are an important energy source in the dairy cow (Herdt, 2000), excessive ketone body formation can result in hyperketonemia (HYK). Defined as a blood BHB concentration ≥ 1.2 mmol/L, HYK is of significant concern to dairy farms as a result of its herd-level incidence in North America of 40% (Duffield, 2000; McArt et al., 2012a), negative effect on cow health and production, and cost to the producer (McArt et al., 2015).

Providing producers with effective, safe, and practical treatment protocols for HYK remains crucial, because the use of nutritional and management strategies for ketosis prevention is not always effective. Early work

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on ketosis therapy suggests that the key principles of treatment should include stimulating gluconeogenesis, increasing utilization of blood glucose, and reducing plasma ketones and free fatty acids (Cote, 1971). A recent systematic review of ketosis therapy performed by Gordon et al. (2013) reiterated these principles. The authors, however, also cited the lack of well-designed ketosis treatment trials and identified oral administration of propylene glycol (PG) as the only HYK therapy for which adequately designed studies were available. Recent studies have evaluated a variety of HYK treatment strategies with minimal effects on health, including dexamethasone as an adjunct therapy to PG; a butaphosphan and cyanocobalamin product and insulin in addition to PG; and daily injections of B₁₂ compared with a negative control (Tatone et al., 2016; Gordon et al., 2017a; Weerathilake et al., 2019). However, Gordon et al. (2017b) did observe a milk yield of 3.1 kg/d more [95% confidence interval (CI) 1.3 to 5.0 kg/d] for cows with a blood glucose concentration <2.2 mmol/L at enrollment when treated with butaphosphan and cyanocobalamin. In another trial, researchers found that treatment with PG plus L-carnitine and methionine improved the resolution of HYK and increased milk yield but had no effect on postpartum health or reproductive performance compared with PG treatment alone (Jeong et al., 2018). As a result of this gap in knowledge related to the effectiveness of adjunct therapies, daily administration of i.v. dextrose and oral PG (alone or in combination) remain commonly used treatments for HYK.

Recently, however, a study comparing PG alone or in combination with i.v. dextrose for treatment of HYK found that cows treated with both i.v. dextrose and oral PG had an improved short-term metabolic response compared with either PG or i.v. dextrose treatment alone, as determined by an increase in insulin concentrations and a decrease in BHB and fatty acid concentrations (Mann et al., 2017). These findings were promising in their support of a combination of PG and i.v. dextrose as a therapy for HYK, but this study was small in scale and not designed to evaluate disease and production outcomes. Accordingly, the authors concluded that additional work was required to fully evaluate i.v. dextrose (500 mL of a 50% solution) as an adjunct therapy to oral PG (300 mL of a 100% solution) to treat HYK.

Our objective was to compare the effects of 3 treatment strategies (oral 100% PG once daily for 3 d; oral 100% PG once daily for 3 d plus i.v. 50% dextrose on the first day of therapy; and oral 100% PG once daily for 3 d plus i.v. 50% dextrose once daily for 3 d) on resolution of HYK, risk of adverse events during the first 60 DIM, and milk yield in early lactation. Our

hypothesis was that the addition of dextrose to a treatment course of 3 d of oral PG would improve recovery, clinical outcomes, and production in early lactation.

MATERIALS AND METHODS

Study Population, Study Design, and Data Collection

We conducted a prospective, randomized controlled trial in Holstein dairy cows on 4 farms (A, B, C, and D) located in central and western New York State from May to September 2019. All farms selected for participation in the trial had to meet the following enrollment criteria: milk at least 1,000 cows; have headlocks in fresh cow pens; use farm management software (Dairy-Comp 305, Valley Agricultural Software, Tulare, CA); and be willing to agree to the proposed testing and treatment protocol. Approval for all animal procedures was granted by the Cornell University Institutional Animal Care and Use Committee (protocol 2019-0054). All farms were asked to sign a consent form agreeing to the proposed testing and treatment protocol.

During the study period, farm A milked 2,100 cows and averaged 40 kg/d of milk per cow; farm B milked 1,000 cows and averaged 40 kg/d; farm C milked 2,500 cows and averaged 38 kg/d; and farm D milked 1,550 cows and averaged 40 kg/d. A controlled energy strategy was used in all 4 herds, and anionic salts were fed to achieve a negative DCAD. All herds milked cows 3 times daily. Cows were uniquely identified using electronic identification, and milk production was measured by on farm milk meters at each milking. Fresh cows were housed in freestall barns with free choice access to water and a farm-specific TMR.

Enrollment occurred once weekly for each herd. Farms A and B enrolled cows on Mondays, farm C on Wednesdays, and farm D on Tuesdays. All animals between 3 and 16 DIM with a previous gestation length of >260 d and no history of retained placenta, metritis, or HYK for the current lactation were eligible for enrollment. Cows were screened for enrollment by study personnel immediately following the morning milking (between 0600 and 0900 h, depending on the farm). Blood samples were obtained from the coccygeal vessels using a 1-mL syringe with a 25-gauge, 1.6-cm needle and were immediately tested for HYK classification using a hand-held ketone meter (Nova Vet Ketone/Glucose Meter; Nova Biomedical, Waltham, MA).

Hyperketonemia was defined as a blood BHB concentration ≥ 1.2 mmol/L, and all HYK-positive animals were assigned to 1 of 3 treatment groups following a randomized block design: (1) 300 mL of a 100% PG solution (100% propylene glycol, USP/EP; Dow, Midland, MI) administered orally once daily for 3 d (**PG3**, n =

121); (2) 300 mL of a 100% PG solution administered orally once daily for 3 d plus 500 mL of a 50% dextrose solution (dextrose 50% injection, VetOne, Boise, ID) administered i.v. on d 1 only (**PG3D1**, $n = 125$); or (3) 300 mL of a 100% PG solution administered orally once daily for 3 d plus 500 mL of a 50% dextrose solution administered i.v. once daily from d 1 through d 3 (**PG3D3**, $n = 127$). All treatments were administered by study personnel. Propylene glycol was administered as a drench using a manual drenching gun (Neogen, Lexington, KY). Cows between 3 and 9 DIM with a BHB concentration <1.2 mmol/L were screened a second time the following week (10–16 DIM) and were eligible for treatment assignment as long as they still met the aforementioned enrollment criteria.

Following treatment, all cows were re-evaluated for post-treatment resolution of HYK (BHB concentration <1.2 mmol/L) 1 wk (7 d) and 2 wk (14 d) after initial HYK diagnosis. This trial had no negative control group because of previous work citing the positive benefits of treating for HYK (McArt et al., 2011, 2012b).

Disease and production data were extracted by study personnel from farm computer records on a weekly basis. Case definitions for the diseases of interest were established by the herd veterinarians and reviewed with farm staff at each site before the beginning of the study to ensure consistency across all study farms. Metritis was diagnosed by farm staff upon the presence of fetid, watery vaginal discharge and hyperthermia (rectal temperature $\geq 39.5^{\circ}\text{C}$). Displaced abomasum was diagnosed by a veterinarian after eliciting a tympanic resonance following auscultation and simultaneous percussion of the intercostal spaces between the 8th and 13th ribs along a diagonal line extending from the olecranon to the tuber coxae of each side of the cow. Farm-diagnosed ketosis was identified by farm personnel using the following criteria: milk deviations based on farm computer software algorithm (farms B and D), no significant findings on physical examination (i.e., displaced abomasum or metritis), and a blood BHB concentration ≥ 1.2 mmol/L (farm C) or a positive urine KetoCheck test (Animal Health, St. Joseph, MO; farm A). Daily milk weights were averaged on a weekly basis to estimate average milk yield (kg/d) over the first 10 wk of lactation.

Sample size estimation was performed using the University of British Columbia Department of Statistics online sample size calculator (www.stat.ubc.ca) via an inference for proportions, comparing 2 independent samples. We based our calculations for a 1-sided hypothesis on the following assumptions: a herd-level HYK prevalence of 30%; an improvement in HYK resolution at 1 wk post-treatment of i.v. dextrose (PG3D1 or PG3D3) over PG alone of 25%; and desired type I

and type II error probabilities of 5 and 20%, respectively. The initial sample size calculation estimated that approximately 120 hyperketonemic cows were required in each of the 3 treatment groups, for a total of 1,320 cows needed for enrollment to account for a potential 10% loss to follow-up. Allocation to treatments within blocks of 3 cows on each farm was performed following a sequence obtained from a random number generator (Research Randomizer v.4.0, Urbaniak and Plous, 2013). Randomization sheets were generated and used by study personnel at the time of treatment allocation. Farm staff and any personnel making subjective assessments were blinded to treatment group.

Our manuscript was written following the REFLECT reporting guidelines for randomized controlled trials in livestock and food safety.

Statistical Analysis

All statistical analyses were performed in SAS version 9.4 (SAS Institute Inc., Cary, NC). Graphs were created using Microsoft Excel (2018; Microsoft Corp., Redmond, WA) and GraphPad Prism version 8.2 (GraphPad, San Diego, CA).

Differences in BHB concentration at initial HYK diagnosis between treatment groups were analyzed using univariable linear regression (PROC MIXED). To improve the normality and homoscedasticity of residuals for blood BHB, we performed natural log-transformation. All geometric least squares means and 95% CI for blood BHB presented hereafter represent the back-transformed values.

To assess the association of treatment group with risk of post-treatment HYK resolution 1 and 2 wk following initial HYK diagnosis and risk of adverse event during the first 60 DIM, we developed explanatory models using fixed-effect multivariable Poisson regression via PROC GENMOD (Ospina et al., 2012). Cows experiencing 1 or more of the following before re-evaluation for post-treatment resolution of HYK at either 1 or 2 wk were not included in the analysis: sold, died, or diagnosed by farm staff with metritis, displaced abomasum, or ketosis. Cows were categorized as having an adverse event if 1 or more of the following occurred on or after the final day of treatment and within the first 60 DIM: sold, died, or diagnosed by farm staff with metritis, displaced abomasum, or ketosis. Additional controlling variables forced into each model regardless of P -value were herd, parity, BHB concentration at initial HYK diagnosis, and study week at initial HYK diagnosis. Biologically plausible pairwise interactions between treatment group and controlling variables were explored if at least 10% of cows were within each level of the contingency table between both variables in

the pairwise interaction; interactions were removed via backward stepwise elimination if $P > 0.05$.

To assess the average daily milk yield during the first 10 wk of lactation, we used repeated-measures ANOVA (PROC MIXED) with week specified in the REPEAT-ED statement. We tested 7 covariance structures (first order autoregressive, compound symmetry, Toeplitz, the heterogeneous form for the aforementioned, as well as unstructured), and we selected unstructured covariance because it yielded the lowest Akaike information criterion. Cows missing milk data for >2 wk were excluded from analysis ($n = 9$) unless they had been removed because they died or were sold. Additional controlling variables forced into all models regardless of P -value were herd, parity, BHB concentration at initial HYK diagnosis, and study week at initial HYK diagnosis. We investigated the interaction of treatment group by week of lactation using a test of simple effects (Winer, 1971), and the interaction was considered important only if all slices were $P \leq 0.05$; no slices met this criteria. We tested other biologically plausible pairwise interactions between treatment group and controlling variables if at least 10% of cows were within each level of the contingency table between both variables in the pairwise interaction and removed via backward stepwise elimination if $P > 0.05$.

For all models, herd was treated as a fixed effect to account for differences in HYK incidences between the 4 farms, and a Tukey–Kramer adjustment was used when controlling for multiple comparisons unless otherwise specified.

RESULTS

Descriptive Statistics

In total, 1,249 cows were screened, and 376 cows (30.1%) were randomly assigned to a treatment group. Incidence of HYK for the 4 herds was as follows: herd A 32.4% ($n = 99$); herd B 30.5% ($n = 67$); herd C 20.3% ($n = 100$); and herd D 47.6% ($n = 110$). Three cows were excluded from analysis for not receiving all 3 d of their respective treatment, resulting in 373 cows eligible for analysis (PG3, $n = 121$; PG3D1, $n = 125$; PG3D3, $n = 127$).

Of the cows eligible for analysis, 64% ($n = 237$) were assigned to a treatment group during the first week of screening, and the remaining 36% ($n = 136$) were assigned during the second week. Week at initial HYK diagnosis was not different between herds ($P = 0.3$) or treatment groups ($P = 0.8$). The median DIM at enrollment were 9 (range 3–16), 7 (range 3–16), and 8 (range: 3–15) for PG3, PG3D1, and PG3D3, respectively. De-

scriptive statistics for treatment groups by herd and parity are shown in Table 1. A histogram of BHB concentrations at initial HYK diagnosis by treatment group is shown in Figure 1. The geometric least squares means of BHB concentration at initial HYK diagnosis were 1.6 mmol/L (95% CI 1.5–1.7 mmol/L), 1.7 mmol/L (95% CI 1.6–1.8 mmol/L), and 1.7 mmol/L (95% CI 1.6–1.8 mmol/L) for PG3, PG3D1, and PG3D3, respectively ($P = 0.2$). The distribution of BHB concentrations at initial HYK diagnosis, 1 wk, and 2 wk following initial HYK diagnosis for each treatment group are shown in Figure 2.

Risk of Hyperketonemia Resolution 1 and 2 wk After Initial Diagnosis

Sixteen cows experienced an adverse event before re-evaluation for post-treatment BHB concentration/HYK resolution at 1 wk following initial HYK diagnosis (PG3, $n = 5$; PG3D1, $n = 2$; PG3D3, $n = 9$), with an additional 4 experiencing an adverse event before 2 wk (PG3, $n = 6$; PG3D1, $n = 4$; PG3D3, $n = 10$). These cows were excluded from subsequent analyses.

Post-treatment hyperketonemia resolution following initial HYK diagnosis for the 3 treatment groups was as follows: at 1 wk, PG3 56.9% ($n = 66$), PG3D1 45.0% ($n = 54$), PG3D3 50.0% ($n = 59$); at 2 wk, PG3 60.0% ($n = 69$), PG3D1 52.1% ($n = 61$), PG3D3 59.5% ($n = 66$). We found no effect of treatment group on the risk of HYK resolution 1 or 2 wk following initial HYK diagnosis, when controlling for the effect of herd, parity, BHB concentration at initial HYK diagnosis, or week at initial HYK diagnosis (both $P = 0.3$; Table 2). The effects of controlling variables at 1 wk following initial HYK diagnosis were as follows: herd, $P = 0.01$; parity, $P = 0.01$; BHB concentration at initial HYK diagnosis, $P < 0.001$; week at initial HYK diagnosis, $P = 0.06$. The effects of controlling variables at 2 wk following initial HYK diagnosis were as follows: herd, $P = 0.001$; parity, $P = 0.001$; BHB concentration at initial HYK diagnosis, $P < 0.001$; week at initial HYK diagnosis, $P = 0.1$.

Risk of Adverse Events

Five cows experienced an adverse event before the final day of treatment (PG3 $n = 2$, PG3D1 $n = 0$, PG3D3 $n = 3$) and were excluded from analysis. The overall incidence of diagnosis with 1 or more adverse events during the first 60 DIM was 9.2% ($n = 34$). The incidence of adverse events within the first 60 DIM for the 3 treatment groups was as follows: PG3 7.6% ($n = 9$), PG3D1 8.0% ($n = 10$), PG3D3 12.1% ($n =$

Table 1. Distribution of treatment groups by herd, parity, and adverse event diagnosis¹

Variable	Treatment group		
	PG3	PG3D1	PG3D3
Herd and week at diagnosis, ² n (% of group)			
A	33 (27.3)	33 (26.4)	33 (26.0)
Wk 1	20 (16.5)	26 (20.8)	21 (16.5)
Wk 2	13 (10.7)	7 (5.6)	12 (9.5)
B	20 (16.5)	23 (18.4)	22 (17.3)
Wk 1	12 (9.9)	13 (10.4)	14 (11.0)
Wk 2	8 (6.6)	10 (8.0)	8 (6.3)
C	32 (26.4)	32 (25.6)	35 (27.6)
Wk 1	25 (20.7)	19 (15.2)	24 (18.9)
Wk 2	7 (5.8)	13 (10.4)	11 (8.7)
D	36 (29.8)	37 (29.6)	37 (29.1)
Wk 1	17 (14.1)	24 (19.2)	22 (17.3)
Wk 2	19 (15.7)	13 (10.4)	15 (11.8)
Parity, n (% of total)			
1	16 (4.3)	21 (5.6)	19 (5.1)
2	32 (8.6)	33 (8.9)	32 (8.6)
≥3	73 (19.6)	71 (19.0)	76 (20.4)
Adverse event, ³ n (% of group)			
One or more ⁴	9 (7.6)	10 (8.0)	15 (12.1)
Metritis	4 (3.4)	2 (1.6)	4 (3.2)
Displaced abomasum	2 (1.7)	3 (2.4)	2 (1.6)
Farm-diagnosed ketosis	0 (0.0)	2 (1.6)	3 (2.4)
Sold or died	4 (3.4)	7 (5.6)	9 (7.3)

¹For 373 Holstein cows administered 300 mL of oral 100% propylene glycol solution for 3 d alone (PG3, n = 121) or in combination with 1 d (PG3D1, n = 125) or 3 d (PG3D3, n = 127) of 500 mL i.v. 50% dextrose for treatment of hyperketonemia on 4 dairy herds in New York State.

²Wk 1 = 3 to 9 DIM. Wk 2 = 10 to 16 DIM.

³Occurring on or after the final day of treatment and within the first 60 DIM (PG3, n = 119; PG3D1, n = 125; PG3D3, n = 124).

⁴Eight cows experienced more than 1 event: 2 cows experienced both metritis and a displaced abomasum, and 6 cows experienced an adverse event before sale or death (metritis, n = 1; displaced abomasum, n = 2; farm-diagnosed ketosis, n = 3).

15). Incidence of metritis, displaced abomasum, farm-diagnosis ketosis, sale, or death can be found in Table 1. A total of 8 cows experienced more than 1 event: 2 cows experienced both metritis and a displaced abomasum, and 6 cows experienced an adverse event before sale or death (metritis n = 1, displaced abomasum n = 2, farm-diagnosed ketosis n = 3). Incidence by herd was 6.1, 9.2, 17.0, and 5.5% for herds A, B, C, and D, respectively.

We found no effect of treatment group on the risk of adverse events during the first 60 DIM when controlling for the effect of herd, parity, BHB concentration at initial HYK diagnosis, or week at initial HYK diagnosis ($P = 0.6$; Table 3). The effects of controlling variables were as follows: herd, $P = 0.4$; parity, $P = 0.007$; BHB concentration at initial HYK diagnosis, $P = 0.08$; week at initial HYK diagnosis, $P = 0.1$. All diagnoses of displaced abomasum were left-sided.

No negative incidents associated with oral PG or i.v. dextrose administration were reported by the study farms.

Milk Production

Average daily milk yield (mean $\pm$ SE) during the first 10 wk of lactation was similar between the 3 treatment groups (PG3 42.7 $\pm$ 0.6 kg/d, PG3D1 42.4 $\pm$ 0.6 kg/d, PG3D3 42.6 $\pm$ 0.6 kg/d; $P = 0.9$; Figure 3). The effect of controlling variables were as follows: week in lactation, $P < 0.001$; herd, $P < 0.001$; parity, $P < 0.001$; BHB concentration at initial HYK diagnosis, $P = 0.002$; and week at initial HYK diagnosis, $P = 0.3$).

DISCUSSION

Oral PG once daily for 3 d has been well established as a beneficial treatment for HYK (McArt et al., 2011; McArt et al., 2012b; Gordon et al., 2013). With recent evidence from Mann et al. (2017) demonstrating a greater decline in BHB concentrations for hyperketonemic cows treated with both PG and dextrose compared to those treated with PG alone, we hypothesized that the addition of dextrose to a treatment course of 3 d

of oral PG would improve HYK recovery, clinical outcomes, and production in early lactation. However, our study found no difference across treatment groups in the resolution of HYK 1 or 2 wk following initial HYK diagnosis, development of adverse events during the first 60 DIM (sale, death, or diagnosed by farm staff with metritis, displaced abomasum, or ketosis) or on milk yield during the first 10 wk of lactation. Furthermore, BHB concentrations remained similar over time regardless of treatment.

The lack of treatment effect in our study was surprising, because findings by Mann et al. (2017) clearly demonstrated a more rapid and sustained reduction in BHB concentrations in the first 3 d following initial diagnosis for cows administered PG in combination with dextrose. However, similar to our study, when Mann et al. (2017) evaluated BHB concentrations at d 7 to 11, they observed no difference between groups. Because we did not sample before 7 d following the initial diagnosis, we cannot know whether the cows in our study also experienced differing BHB dynamics between treatment groups. Furthermore, because Mann et al. (2017) did not evaluate health and production parameters, we cannot know whether the treatment effect they observed would have translated to clinically important outcomes.

Interestingly, the lack of treatment effect in our study appeared to be consistent regardless of BHB concentration at initial HYK diagnosis. Similar to previous research (McArt et al., 2012a; Tatone et al., 2016), cows with higher BHB concentrations at initial HYK diagnosis tended to have an increased risk of adverse event

diagnosis and demonstrated decreased milk production. Resolution of HYK was also negatively affected by BHB concentration at the time of initial diagnosis. However, we observed no interaction between BHB concentration at initial diagnosis and any outcome of interest in our analysis. It is important to note that these findings must be interpreted with caution for cows with BHB concentrations ≥ 3.0 mmol/L, because $<10\%$ of cows in each treatment group had BHB concentrations within this range at initial diagnosis (Figure 1). Given that dextrose is commonly recommended in treatment protocols for cows with BHB concentrations ≥ 3.0 mmol/L, additional research is advised to determine whether it truly is advantageous to this population of hyperketonemic cows.

Herd incidence of HYK also did not influence the lack of treatment effect. The overall incidence of HYK in this study was 30.1%, but herd incidences ranged from 20.3 to 47.6%. Previous reports of HYK incidence in North American cite rates upwards of 40 to 60% (Duffield et al., 1998; McArt et al., 2012b), meaning that our herds represented low and average incidences. Nevertheless, we observed no treatment $\times$ herd interaction, suggesting that our findings may be relevant to herds with a wide range of HYK incidences.

Of all cows randomly assigned to a treatment in this study, 64% were first diagnosed with HYK during the first week of screening (3–9 DIM) and 36% during the second week (10–16 DIM). This was in line with previous reports that showed peak incidence of HYK at 5 DIM (McArt et al., 2012a). However, week of diagnosis did not appear to affect any of our outcomes of interest.

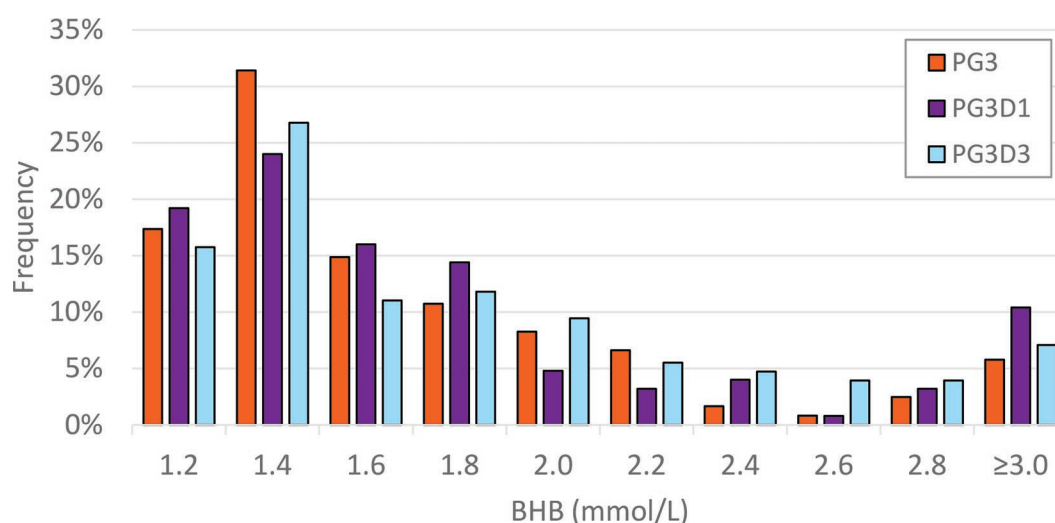


Figure 1. Histogram showing BHB concentrations at initial hyperketonemia diagnosis by treatment group for 373 cows administered 300 mL of a 100% oral propylene glycol solution for 3 d alone (PG3; orange bars) or in combination with 1 d (PG3D1; purple bars) or 3 d (PG3D3; light blue bars) of 500 mL i.v. 50% dextrose for treatment of hyperketonemia.

Table 2. Final Poisson regression model outcomes evaluating the association of treatment group with resolution of hyperketonemia at 1 and 2 wk following initial hyperketonemia diagnosis^{1,2}

Time and treatment group	n (%) ³	Estimate	SE	P-value ⁴	Relative risk	95% CI ⁴
1 wk post-diagnosis ⁵						
PG3	66 (56.9)	Referent	—	—	—	—
PG3D1	54 (45.0)	−0.19	0.13	0.3	0.8	0.6–1.1
PG3D3	59 (50.0)	−0.07	0.13	0.9	0.9	0.7–1.3
2 wk post-diagnosis ⁶						
PG3	69 (60.0)	Referent	—	—	—	—
PG3D1	61 (52.1)	−0.12	0.12	0.5	0.9	0.7–1.2
PG3D3	66 (59.5)	0.06	0.11	0.8	1.1	0.8–1.4

¹For 373 Holstein cows administered 300 mL of oral 100% propylene glycol solution for 3 d alone (PG3) or in combination with 1 d (PG3D1) or 3 d (PG3D3) of 500 mL i.v. 50% dextrose for treatment of hyperketonemia on 4 dairy herds in New York State.

²Hyperketonemia resolution defined as treated cows with BHB concentration <1.2 mmol/L.

³For resolution within group.

⁴P-value and 95% CI reported for Tukey's comparison.

⁵Total n = 354 (PG3, n = 116; PG3D1, n = 120; PG3D3, n = 118). Variables included in the model: treatment group ($P = 0.3$), herd ($P = 0.01$), parity ($P = 0.01$), BHB concentration at initial hyperketonemia diagnosis ($P < 0.001$), week at initial hyperketonemia diagnosis ($P = 0.06$).

⁶Total n = 343 (PG3, n = 115; PG3D1, n = 117; PG3D3, n = 111). Variables included in the model: treatment group ($P = 0.3$), herd ($P = 0.001$), parity ($P = 0.001$), BHB concentration at initial hyperketonemia diagnosis ($P < 0.001$), week at initial hyperketonemia diagnosis ($P = 0.1$).

Because our study was not designed to examine this relationship, it is most likely that power limited our ability to elucidate this.

The proportion of animals experiencing HYK resolution in this study ranged from 45.9 to 59.7% at 1 wk following initial HYK diagnosis, and 53.7 to 61.2%

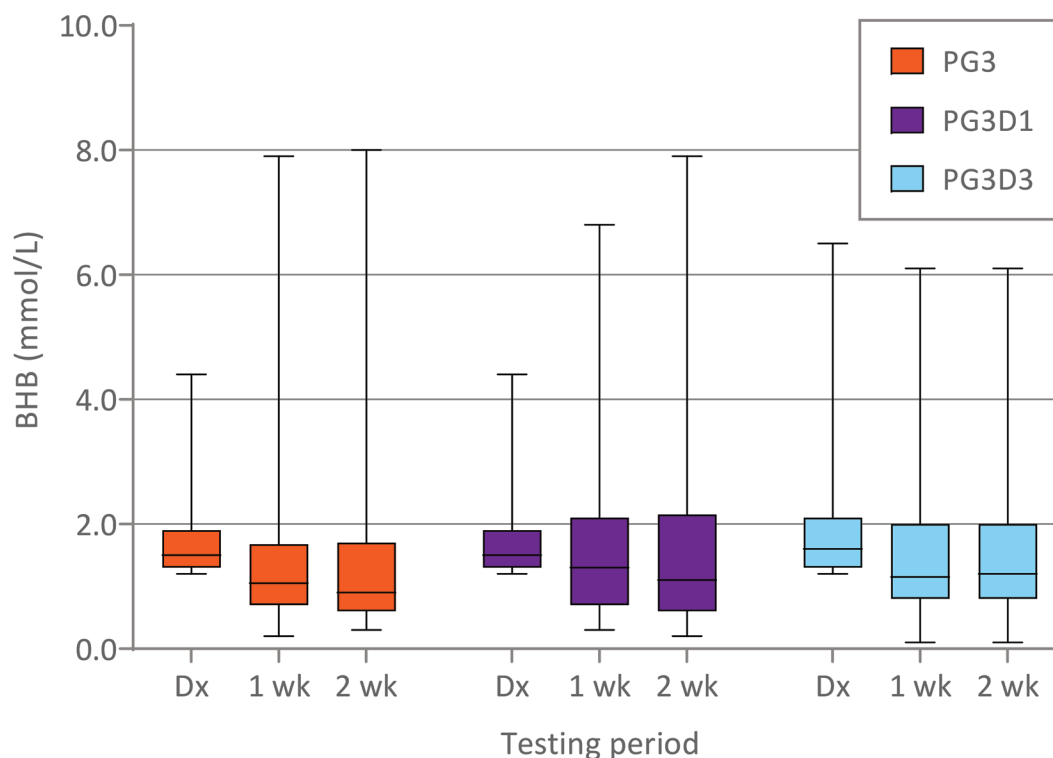


Figure 2. Distribution of BHB concentrations for each treatment group at initial hyperketonemia diagnosis (Dx), and 1 and 2 wk following initial hyperketonemia diagnosis, for 373 cows administered 300 mL of oral 100% propylene glycol solution for 3 d alone (PG3; n = 121; orange bars) or in combination with 1 d (PG3D1; n = 125; purple bars) or 3 d (PG3D3; n = 127; light blue bars) of 500 mL i.v. 50% dextrose for treatment of hyperketonemia. The box and whiskers represent the median and interquartile range, and the minimum and maximum observations, respectively.

Table 3. Final Poisson regression model outcomes evaluating the association of treatment group with occurrence of adverse event diagnosis within the first 60 DIM^{1,2}

Treatment group ³	n (%) ⁴	Estimate	SE	P-value ⁵	Relative risk	95% CI ⁵
PG3	9 (7.6)	Referent	—	—	—	—
PG3D1	10 (8.0)	0.18	0.45	0.9	1.2	0.4 to 3.5
PG3D3	15 (12.1)	0.40	0.42	0.6	1.5	0.6 to 4.0

¹For 373 Holstein cows administered 300 mL of oral 100% propylene glycol solution for 3 d alone (PG3) or in combination with 1 d (PG3D1) or 3 d (PG3D3) of 500 mL i.v. 50% dextrose for treatment of hyperketonemia on 4 dairy herds in New York State.

²Cows were categorized as having an adverse event if one or more of the following occurred on or after the day of sampling and within the first 60 DIM: sold, died, or diagnosed by farm staff with metritis, displaced abomasum, or ketosis. Eight cows experienced more than 1 event: 2 cows experienced both metritis and a displaced abomasum, and 6 cows experienced an adverse event before being sold or dying (metritis, n = 1; displaced abomasum, n = 2; farm-diagnosed ketosis, n = 3).

³Total n = 368 (PG3, n = 119; PG3D1, n = 125; PG3D3, n = 124). Variables included in the model: treatment group ($P = 0.6$), herd ($P = 0.4$), parity ($P = 0.007$), BHB concentration at initial hyperketonemia diagnosis ($P = 0.08$), week at initial hyperketonemia diagnosis ($P = 0.1$).

⁴For adverse event diagnosis within group.

⁵P-value and 95% CI reported for Tukey's comparison.

at 2 wk. These findings were consistent with previous research on the resolution of HYK 1 wk following treatment, which demonstrated approximately 60% resolution (McArt et al., 2011; Gordon et al., 2013). We theorize that resolution may not have been greater at 2 wk following initial HYK diagnosis in part because of the range of BHB concentrations observed: some cows experienced BHB concentrations upwards of 6.5 mmol/L. Cows with high concentrations need time for levels to drop to < 1.2 mmol/L, and 2 wk may not have been sufficient. However, as previously mentioned, <10% of cows in each treatment group fell into this

high BHB category, suggesting that time does not fully explain the lack of resolution. Continued work in understanding this non-resolved population is needed to determine whether additional treatment is warranted or whether this is a subpopulation of cows that may be efficient at mobilizing and using BHB to support higher milk production (Duffield et al., 2009).

We cannot ignore the potential that our findings could be a result of a type II error. Initial sample size calculations included a suspected increase in HYK resolution at 1 wk following initial HYK diagnosis for either of the dextrose-treated groups of 25% over those treated with

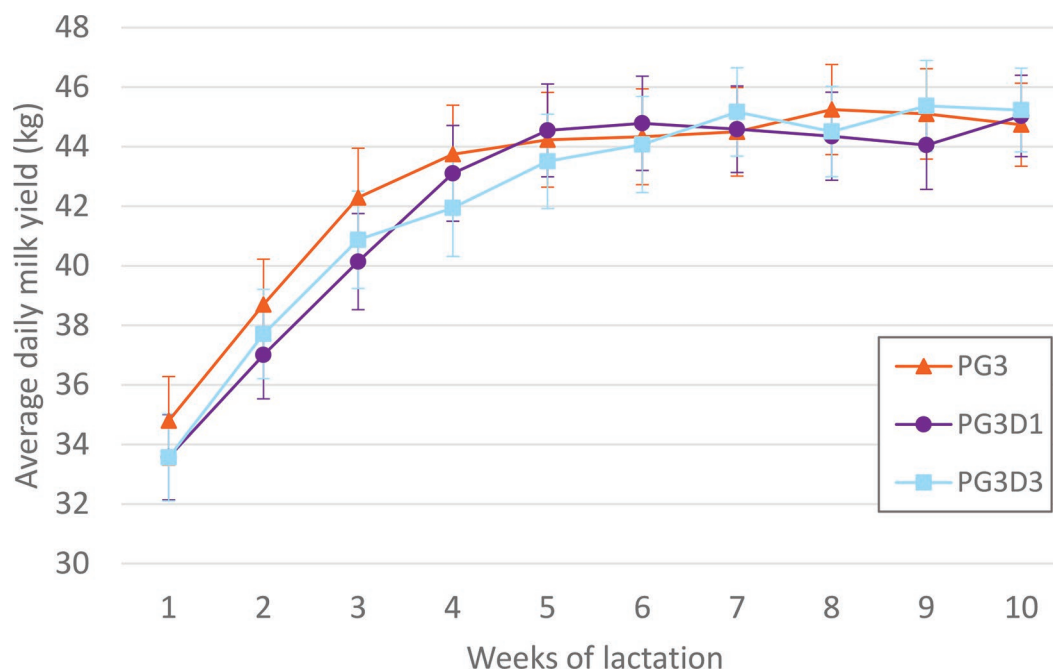


Figure 3. Average daily milk yield during the first 10 wk of lactation for 364 cows administered 300 mL of oral 100% propylene glycol solution for 3 d alone (PG3; n = 121; orange triangles) or in combination with 1 d (PG3D1; n = 125; purple circles) or 3 d (PG3D3; n = 127; light blue squares) of 500 mL i.v. 50% dextrose for treatment of hyperketonemia. Error bars represent 95% confidence intervals.

PG alone. Therefore, our sample size should have been sufficient to elucidate a large difference in HYK resolution between treatment groups if one truly existed, but it may not have been great enough to reveal a small difference. Nonetheless, it can be argued that small differences are biologically inconsequential, because the added labor and cost of i.v. dextrose administration might outweigh the minimal benefit. Furthermore, the fact that the lack of a treatment effect was consistent across all BHB concentrations at diagnosis and across all 4 herds (i.e., we found no treatment interaction effect for any of our models), suggests that this lack of effect may be real, at least in the case of HYK resolution.

Unfortunately, our study was likely underpowered to draw a definitive conclusion regarding a difference in adverse events or milk production between treatment groups. One limitation of our study was the relatively low incidence of adverse events compared with other published data (McArt et al., 2012a; Ribeiro et al., 2013). However, incidence of adverse events did vary among herds included in this study (5.5 to 17.0%). Assuming an adverse event incidence for hyperketonemic cows of 15%, at our initial sample size and power, we would be able to identify a difference only between groups of $\geq 10\%$ points. Likewise, assuming a mean milk production of 40 kg/d (SD = 6 kg/d), differences in milk production would have to be ≥ 2 kg/d to be detected.

CONCLUSIONS

Hyperketonemia continues to be a significant disease, both in prevalence and cost. Providing dairies with valid, labor efficient, and cost-effective strategies for treating this disease is important. Our study demonstrates that the addition of i.v. dextrose as an adjunct therapy to oral PG provided no improvement in resolution of HYK, no reduction in adverse events, and no increase in average daily milk yield over treatment with PG alone. Administering dextrose i.v. is labor intensive and has the potential side effect of jugular thrombophlebitis from extravascular administration. Because our study demonstrates no advantage of the addition of dextrose to PG therapy for the treatment of HYK, discontinuing its use as an adjunct therapy to oral PG might allow for investigation of a less invasive treatment option with no negative effect on treatment success.

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REFERENCES

- Baird, G. D. 1982. Primary ketosis in the high-producing dairy cow: Clinical and subclinical disorders, treatment, prevention, and outlook. *J. Dairy Sci.* 65:1–10. [https://doi.org/10.3168/jds.S0022-0302\(82\)82146-2](https://doi.org/10.3168/jds.S0022-0302(82)82146-2).
- Bauman, D. E., and W. B. Currie. 1980. Partitioning of nutrients during pregnancy and lactation: A review of mechanisms involving homeostasis and homeorhesis. *J. Dairy Sci.* 63:1514–1529. [https://doi.org/10.3168/jds.S0022-0302\(80\)83111-0](https://doi.org/10.3168/jds.S0022-0302(80)83111-0).
- Chow, S.-C., J. Shao, H. Wang, and Y. Lokhnygina. 2017. *Sample Size Calculations in Clinical Research*. 3rd ed. Chapman and Hall/CRC, London, UK.
- Cote, J. F. 1971. Bovine ketosis: Principles of therapy. *Can. Vet. J.* 12:19–20.
- Duffield, T. 2000. Subclinical ketosis in lactating dairy cattle. *Vet. Clin. North Am. Food Anim. Pract.* 16:231–253. [https://doi.org/10.1016/S0749-0720\(15\)30103-1](https://doi.org/10.1016/S0749-0720(15)30103-1).
- Duffield, T. F., K. D. Lissemore, B. W. McBride, and K. E. Leslie. 2009. Impact of hyperketonemia in early lactation dairy cows on health and production. *J. Dairy Sci.* 92:571–580. <https://doi.org/10.3168/jds.2008-1507>.
- Duffield, T. F., D. Sandals, K. E. Leslie, K. Lissemore, B. W. McBride, J. H. Lumsden, P. Dick, and R. Bagg. 1998. Efficacy of monensin for the prevention of subclinical ketosis in lactating dairy cows. *J. Dairy Sci.* 81:2866–2873. [https://doi.org/10.3168/jds.S0022-0302\(98\)75846-1](https://doi.org/10.3168/jds.S0022-0302(98)75846-1).
- Gordon, J. L., T. F. Duffield, T. H. Herdt, D. F. Kelton, L. Neuder, and S. J. LeBlanc. 2017a. Effects of a combination butaphosphan and cyanocobalamin product and insulin on ketosis resolution and milk production. *J. Dairy Sci.* 100:2954–2966. <https://doi.org/10.3168/jds.2016-11925>.
- Gordon, J. L., S. J. LeBlanc, and T. F. Duffield. 2013. Evaluation of accuracy of an electronic beta-hydroxybutyrate meter using fresh and stored whole blood and serum from dairy cows. *J. Dairy Sci.* 96:55–56.
- Gordon, J. L., S. J. LeBlanc, D. F. Kelton, T. H. Herdt, L. Neuder, and T. F. Duffield. 2017b. Randomized clinical field trial on the effects of butaphosphan-cyanocobalamin and propylene glycol on ketosis resolution and milk production. *J. Dairy Sci.* 100:3912–3921. <https://doi.org/10.3168/jds.2016-11926>.
- Herdt, T. H. 2000. Ruminant adaptation to negative energy balance. Influences on the etiology of ketosis and fatty liver. *Vet. Clin. North Am. Food Anim. Pract.* 16:215–230. [https://doi.org/10.1016/S0749-0720\(15\)30102-X](https://doi.org/10.1016/S0749-0720(15)30102-X).
- Jeong, J. K., I. S. Choi, S. H. Moon, S. C. Lee, H. G. Kang, Y. H. Jung, S. B. Park, and I. H. Kim. 2018. Effect of two treatment protocols for ketosis on the resolution, postpartum health, milk yield, and reproductive outcomes of dairy cows. *Theriogenology* 106:53–59. <https://doi.org/10.1016/j.theriogenology.2017.09.030>.
- Mann, S., F. A. Leal Yepes, E. Behling-Kelly, and J. A. A. McArt. 2017. The effect of different treatments for early-lactation hyperketonemia on blood β -hydroxybutyrate, plasma nonesterified fatty

- acids, glucose, insulin, and glucagon in dairy cattle. *J. Dairy Sci.* 100:6470–6482. <https://doi.org/10.3168/jds.2016-12532>.
- McArt, J. A. A., D. V. Nydam, and G. R. Oetzel. 2012a. Epidemiology of subclinical ketosis in early lactation dairy cattle. *J. Dairy Sci.* 95:5056–5066. <https://doi.org/10.3168/jds.2012-5443>.
- McArt, J. A. A., D. V. Nydam, and G. R. Oetzel. 2012b. A field trial on the effect of propylene glycol on displaced abomasum, removal from herd, and reproduction in fresh cows diagnosed with subclinical ketosis. *J. Dairy Sci.* 95:2505–2512. <https://doi.org/10.3168/jds.2011-4908>.
- McArt, J. A. A., D. V. Nydam, P. A. Ospina, and G. R. Oetzel. 2011. A field trial on the effect of propylene glycol on milk yield and resolution of ketosis in fresh cows diagnosed with subclinical ketosis. *J. Dairy Sci.* 94:6011–6020. <https://doi.org/10.3168/jds.2011-4463>.
- McArt, J. A. A., D. V. V. Nydam, and M. W. W. Overton. 2015. Hyperketonemia in early lactation dairy cattle: A deterministic estimate of component and total cost per case. *J. Dairy Sci.* 98:2043–2054. <https://doi.org/10.3168/jds.2014-8740>.
- Ospina, P. A., D. V. Nydam, and T. J. DiCiccio. 2012. Technical note: The risk ratio, an alternative to the odds ratio for estimating the association between multiple risk factors and a dichotomous outcome. *J. Dairy Sci.* 95:2576–2584. <https://doi.org/10.3168/jds.2011-4515>.
- Ribeiro, E. S., F. S. Lima, L. F. Greco, R. S. Bisinotto, A. P. A. Monteiro, M. Favoreto, H. Ayres, R. S. Marsola, N. Martinez, W. W. Thatcher, and J. E. P. Santos. 2013. Prevalence of periparturient diseases and effects on fertility of seasonally calving grazing dairy cows supplemented with concentrates. *J. Dairy Sci.* 96:5682–5697. <https://doi.org/10.3168/jds.2012-6335>.
- Tatone, E. H., T. F. Duffield, M. B. Capel, T. J. DeVries, S. J. LeBlanc, and J. L. Gordon. 2016. A randomized controlled trial of dexamethasone as an adjunctive therapy to propylene glycol for treatment of hyperketonemia in postpartum dairy cattle. *J. Dairy Sci.* 99:8991–9000. <https://doi.org/10.3168/jds.2016-11358>.
- Urbaniak, G. C., and S. Plous. 2013. Research Randomizer (Version 4.0) Computer Software. Accessed May 23, 2019. <http://www.randomizer.org>.
- Weerathilake, W. A. D. V., A. H. Brassington, S. J. Williams, W. Y. Kwong, L. A. Sinclair, and K. D. Sinclair. 2019. Added dietary cobalt or vitamin B12, or injecting vitamin B12 does not improve performance or indicators of ketosis in pre- and post-partum Holstein-Friesian dairy cows. *Animal* 13:750–759. <https://doi.org/10.1017/S175173111800232X>.
- Winer, B. J. 1971. *Statistical Principles in Experimental Design*. 2d Ed. McGraw-Hill, New York, NY.

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